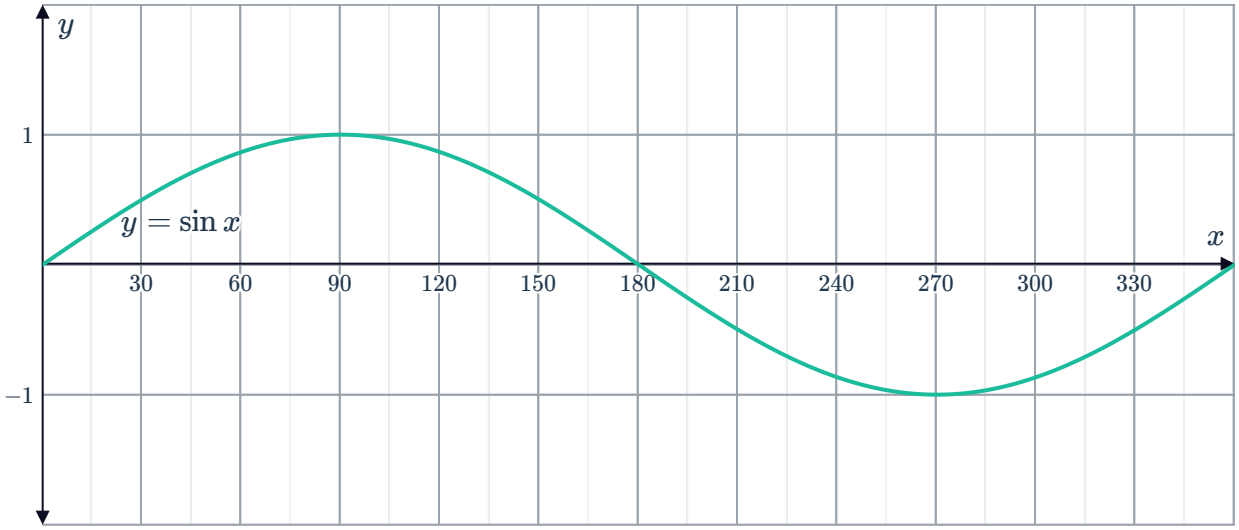
Sine function

1

a

x	0°	30°	90°	150°	180°	210°	270°	330°	360°
$\sin x$	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0

b



c Negative

d Quadrant 4

e The function values and shape of the graph repeat at regular intervals.

f Period: 360°

g $(0, 0)$

h 1

i -1

j $[-1, 1]$

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2

a

x	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{3\pi}{2}$	$\frac{11\pi}{6}$	2π
$\sin x$	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0

b



c Period: 2π

3

a Because the range of the unit circle is $-1 \leq y \leq 1$.

b Because the unit circle repeats every 360°.

4

a Yes **b** No **c** Yes **d** No

e Yes **f** No **g** Yes

5

a

i Decreasing **ii** Increasing **iii** Decreasing **iv** Increasing

b

i 180° **ii** -180°

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6

a

i Decreasing **ii** Increasing **iii** Decreasing **iv** Increasing

b

i π **ii** $-\pi$

7

a $50\,000$

b The peak number of visitors to Times Square each day is $50\,000$.

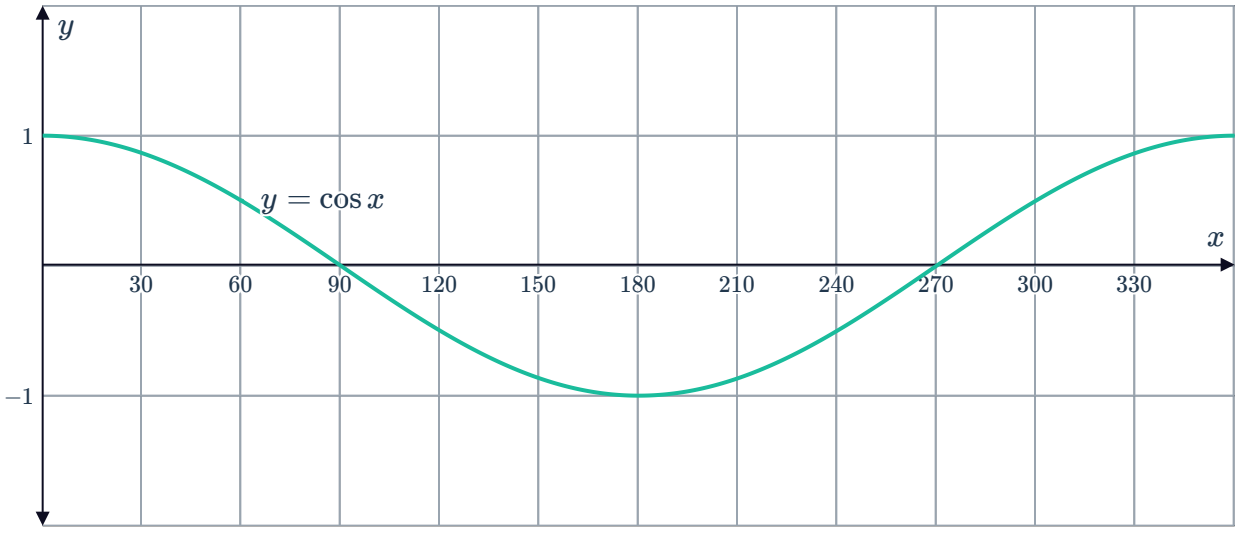
Cosine function

8

a

x	0°	60°	90°	120°	180°	240°	270°	300°	360°
$\cos x$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1

b



c Positive

d Quadrant 4

e $(0, 1)$

f 1

g -1

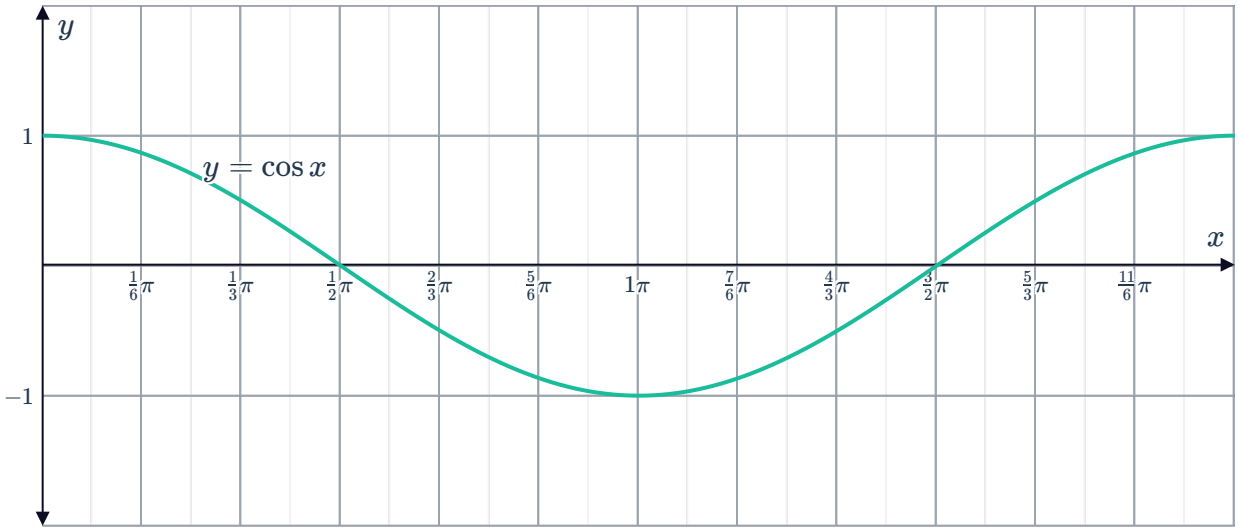
h $-\infty < x < \infty$

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9 **a**

x	0	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	2π
$\cos x$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1

b



10

a Because the domain of the unit circle is $-1 \leq y \leq 1$.

b Because the unit circle repeats every 360°.

11

a Yes **b** No **c** No **d** Yes

e No **f** No **g** Yes

12

a 270°

b

i Increasing **ii** Decreasing **iii** Decreasing **iv** Increasing

c

i $90^\circ, 270^\circ$ **ii** $-270^\circ, -90^\circ$

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13

a $\frac{3\pi}{2}$

b

i Increasing **ii** Decreasing **iii** Decreasing **iv** Increasing

c

i $\frac{\pi}{2}, \frac{3\pi}{2}$ **ii** $-\frac{3\pi}{2}, -\frac{\pi}{2}$

14

a No **b** Yes **c** No **d** No

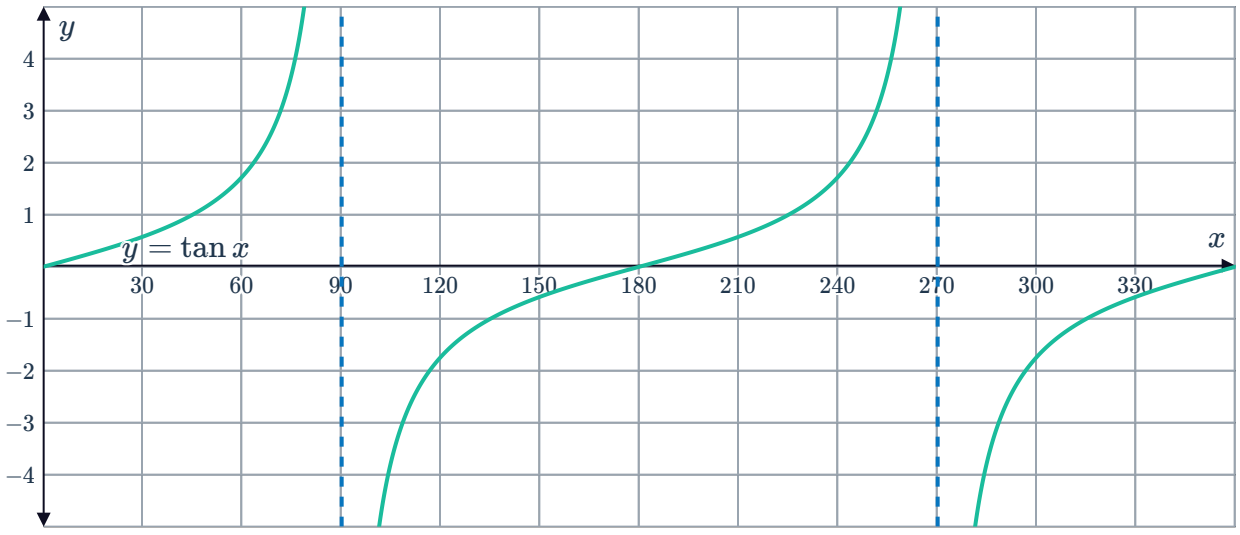
Tangent function

15

a

x	0°	60°	90°	120°	135°	180°	225°	270°	300°	360°
$\tan x$	0	$\sqrt{3}$	–	$-\sqrt{3}$	-1	0	1	–	$-\sqrt{3}$	0

b



c $(0, 0)$

d Negative

e Quadrant 4

16

a Yes **b** No **c** No **d** Yes

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17

a 180° **b** $-\infty < y < \infty$ **c** 630°

18

a $y = 0$ **b** π **c** $x = \frac{\pi}{2}, \frac{3\pi}{2}$ **d** Increase

e $x = \pi$

19

a π **b** $-2\pi, -\pi, 0, \pi, 2\pi$

c $\frac{\pi}{4}$ **d** $\frac{5\pi}{4}, -\frac{3\pi}{4}, -\frac{7\pi}{4}$

e $\frac{3\pi}{4}, \frac{7\pi}{4}, \frac{-\pi}{4}, \frac{-5\pi}{4}$

20

a The opposite side increases.

b The value of $\tan\theta$ increases.

c $\tan\theta = \frac{\sin\theta}{\cos\theta}$

d $\theta = -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$

e $\theta = -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$

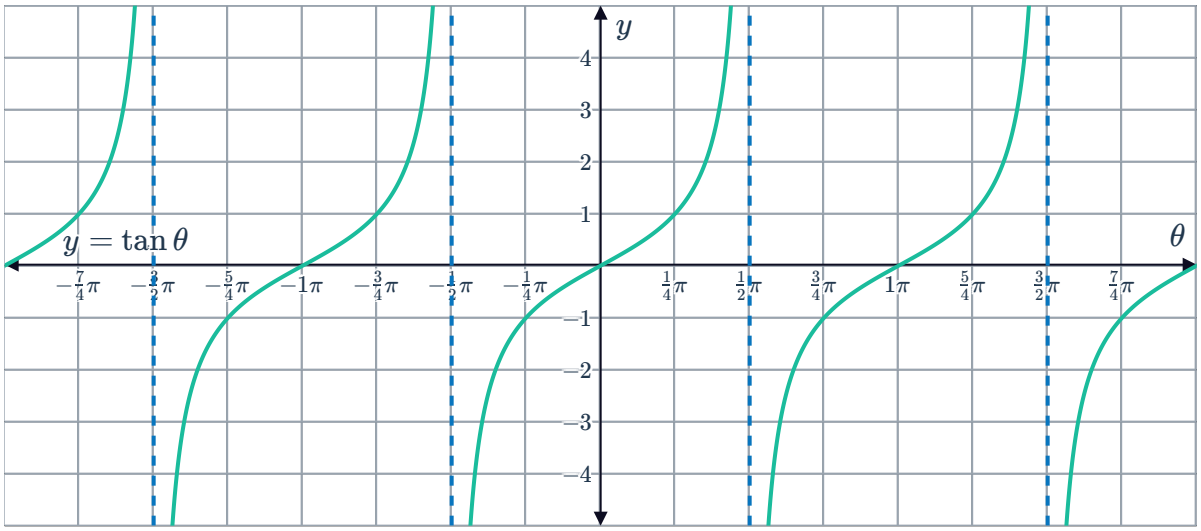
f

θ	-2π	$-\frac{7\pi}{4}$	$-\frac{5\pi}{4}$	$-\pi$	$-\frac{3\pi}{4}$	$-\frac{\pi}{4}$
$\tan\theta$	0	1	-1	0	1	-1

θ	0	$\frac{\pi}{4}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{7\pi}{4}$	2π
$\tan\theta$	0	1	-1	0	1	-1	0

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g



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